Major Project Idea: Centralized Behavioral Drift Detection System

Manraj Bhullar

British Columbia Institute of Technology

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Instructor: D'Arcy Smith

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The Tagline

Detect encrypted command-and-control traffic by analyzing behavioral drift from established communication baselines across hosts.

The Story

A security analyst is responsible for monitoring several Linux servers that support a range of applications, services, and infrastructure workloads. The servers regularly communicate with other hosts and most communication is encrypted. During normal monitoring, network activity appears consistent and does not trigger alerts.

During a non-routine audit, the analyst notices that multiple hosts have begun establishing encrypted sessions with unfamiliar remote destinations that do not match their normal communication behavior. The connections use standard ports and appear legitimate when viewed individually. After reviewing communication logs across systems, the analyst discovers communication behavior shifted gradually over time and noticed traffic patterns consistent with automated command-and-control behavior.

The activity remained undetected because most monitoring tools evaluate communication events in isolation and focus on identifying known malicious indicators rather than maintaining behavioral baselines over time. Many tools also rely on deep packet inspection, which loses visibility when traffic is encrypted. While effective at detecting known threats, these systems often do not continuously and explicitly compare current activity against historical baseline behavior. Encrypted sessions with predictable timing can appear legitimate when viewed individually, allowing suspicious behavior to develop gradually as communication patterns shift.

This highlights the need for monitoring that establishes baselines and continuously evaluates changes to detect behavioral drift across hosts.

The Solution

This project builds a centralized monitoring system supported by lightweight Linux host agents. The agents collect encrypted communication metadata and send it to a central controller that builds baseline communication profiles for each monitored host and evaluates patterns across them. The controller continuously compares new communication activity against established baselines to detect behavioral drift. When drift exceeds defined thresholds, the system generates alerts and presents response actions such as restricting communication or blocking suspicious destinations. The system is designed to monitor communication behavior rather than payload contents, allowing it to detect suspicious activity even when traffic is encrypted.

The Hook (One Sentence)

A centralized monitoring system that detects behavioral drift in encrypted network communication across Linux hosts by continuously comparing communication activity against established behavioral baselines.

The Spoken Summary (One Paragraph)

Traditional monitoring tools often struggle to detect malicious activity when network traffic is encrypted or when threats develop gradually through small communication changes. This project builds a centralized monitoring system using lightweight Linux host agents that collect encrypted communication metadata and establish behavioral baselines for each monitored host. A central controller continuously compares new communication activity against these baselines to identify behavioral drift associated with command-and-control activity. The system generates alerts and provides response options that require manual approval through a centralized dashboard. The project is designed to operate within a controlled Linux lab environment and its performance will be evaluated using reproducible communication scenarios and tests.

The Core Plan (One Page)

Modern networking relies heavily on encrypted communication, which increases the importance of monitoring how communication behavior changes over time. Malicious activity often appears through gradual shifts in how often systems communicate, where they connect, or how much data they exchange rather than single-event indicators. Monitoring these shifts requires systems that establish behavioral baselines and continuously compare new communication activity against those baselines.

This project builds a centralized monitoring system designed to detect behavioral drift across distributed Linux hosts. Each monitored host runs a lightweight worker agent that collects encrypted communication details, including when connections happen, where they connect, how long they last, and how much data is transferred within sessions. The collected information is sent to a central controller that builds and maintains baseline communication profiles for each monitored host, while also correlating behaviour across hosts. The controller continuously compares new communication activity against these baselines to identify deviations that may indicate command-and-control behavior.

The system identifies behavioral drift by monitoring three primary communication behaviors. These include the introduction of communication with new or previously unknown destinations, changes in how frequently communication occurs, and changes in the amount of data exchanged during communication. Monitoring these behaviors allows the system to detect gradual changes in encrypted communication patterns while maintaining a focused and measurable detection scope.

When behavioral drift exceeds defined thresholds, the system generates alerts through a centralized dashboard. The dashboard provides visibility into baseline comparisons and presents response options, including blocking communication from suspicious destinations or temporarily restricting network communication from affected hosts. All response actions are designed to be logged and reversible.

The system will be demonstrated using multiple Linux hosts in a controlled lab environment. Evaluation will be performed using simulated network activity that represents normal communication and drift behavior. Project success will be measured using detection accuracy, alert response time, and data collection consistency. The project intentionally limits monitoring to encrypted communication metadata and baseline comparison logic to ensure feasibility and clear evaluation within one academic term.

The Full Details (Three Pages)

The Problem and Why It Matters

Network monitoring systems traditionally rely on identifying known malicious indicators or analyzing individual communication events. As encrypted communication has become the standard for modern networking, visibility into packet payload contents is reduced, making it harder to detect malicious activity using traditional techniques. Attackers increasingly rely on encrypted communication channels and automated command-and-control techniques that blend into legitimate network traffic.

Compromised systems often demonstrate gradual behavioral changes rather than immediate or obvious malicious actions. These changes can include introducing communication with unknown destinations, increasing how frequently communication occurs, or exchanging abnormal amounts of data during communication sessions. When viewed individually, these changes may appear legitimate and fail to trigger traditional monitoring alerts. Detecting these gradual changes could be solved by a monitoring system that establishes behavioral baselines and continuously evaluates how communication activity changes over time and correlating it across multiple hosts.

The Gap in Current Approaches

Current command-and-control detection methods commonly focus on identifying known malicious destinations, suspicious domains, or communication patterns that match known attack signatures. These methods are effective when attackers reuse known attack patterns that are recognizable, however, they can miss command-and-control activity when attackers use encrypted communication, legitimate services, or frequently changing destinations. In these situations, individual communication sessions often appear normal and do not trigger alerts, even though communication behavior may be gradually changing across systems and may only become noticeable when viewed collectively.

This project focuses specifically on detecting command-and-control behavior by monitoring how communication patterns change over time using baseline comparison across hosts. The system is intentionally limited to analyzing encrypted communication metadata, including connection timing, destination changes, and communication volume. It does not inspect packet contents, decrypt traffic, or attempt to detect all types of network threats. Instead, it targets gradual behavioral drift that may indicate command-and-control activity while maintaining a narrow and measurable detection scope. This type of monitoring is designed to serve as a supplemental and specialized detection layer that could possibly integrate or work in tandem with larger security monitoring platforms.

Solution Overview

The proposed system consists of distributed agents and a centralized controller for monitoring. Each monitored Linux host runs a lightweight worker agent that collects encrypted communication details, including when connections occur, where they connect, how long they remain active, and how much data is transferred. The worker agents transmit this information to a central controller responsible for establishing and maintaining baseline communication behavior for each monitored host and analyzing patterns across hosts.

Drift Categories

The controller continuously compares new communication activity against established baseline behavior to detect behavioral drift. Detection is based on three preliminary drift categories (subject to change):

- **New Destination Drift:** Triggers when a host communicates with a destination that it unfamiliar.
- **Connection Frequency Drift:** Triggers when connection frequency changes beyond predefined thresholds relative to baseline activity.
- **Traffic Volume Drift:** Triggers when the amount of data exchanged during communication deviates from baseline behavior.

When drift is detected, alerts are generated through a centralized dashboard that provides baseline comparison details and response options.

Response and Safety Controls

The system will provide two approved mitigation options:

- Blocking communication from suspicious destinations
- Temporarily restricting network communication from affected hosts

All mitigation actions require manual approval and support rollback functionality to return to normal operation.

Scope Limits

In Scope

- Monitoring Linux hosts
- Collecting encrypted communication metadata
- Establishing baseline communication behavior
- Detecting drift across three defined communication categories
- Centralized monitoring dashboard
- Response actions
- Evaluation using simulated network activity

Out of Scope

- Packet payload inspection or traffic decryption
- Machine learning based detection
- Multi-operating system endpoint support
- Enterprise SIEM integration
- Production scale deployment

Success Metrics

Detection Accuracy: Measures the system's ability to correctly identify drift behavior.

- *Test Method:* Simulated network activity representing normal and drift communication behavior
- *Pass Threshold:* Detect at least 90% of drift events with no more than one false alert per host.

Alert Response Time: Measures how quickly alerts are generated after drift behavior occurs.

- *Test Method:* Monitor system dashboard after running simulated test.
- *Pass Threshold:* Alerts generated within 30 seconds of drift detection during testing scenarios.

Data Collection Consistency: Measures the reliability of communication data collected by worker agents.

- *Test Method:* Continuous monitoring over a 24-hour testing period.
- *Pass Threshold*: At least 99% of expected communication records are successfully collected and processed,

Risks and Mitigations

Inaccurate Baseline: Normal system behavior changes may be incorrectly identified as behavioral drift.

- *Mitigation:* Monitor and evaluate conditions under normal behavior and adjust drift detection thresholds and logic accordingly.

Inaccurate Data Collection: Incomplete or inconsistent communication data may reduce detection accuracy.

- *Mitigation:* Validate collected data using secondary tracking tools.

Deployment Complexity: Monitoring multiple hosts may increase configuration and coordination complexity early on.

- *Mitigation:* Develop and validate an early single-host prototype before expanding to multi-host monitoring.

Mitigation Uncertainty: Response actions may unintentionally disrupt legitimate communication.

- *Mitigation*: Require manual approval for response actions and maintain rollback functionality.

Feasibility Within One Academic Term

The project is constrained to monitoring encrypted communication metadata and rule-based baseline comparison within a controlled Linux lab environment.

Development will follow four stages:

- Worker agent data collection and centralized storage
- Baseline generation and drift detection logic
- Alert dashboard and mitigation workflow
- Simulated activity testing and performance validation

Biography (Student Feasibility Statement)

I am currently completing a Bachelor of Science in Applied Computer Science at the British Columbia Institute of Technology with a specialization in Network Security Applications Development. My academic experience includes Linux system administration, network socket programming, and coursework that involved implementing a distributed controller-worker assignment. Through this coursework, I have worked with network communication protocols and system-level programming concepts that relate directly to distributed monitoring and communication analysis.

I have also completed network monitoring labs and coursework that involved analyzing communication behavior and network activity patterns within controlled environments. These experiences support the data collection, monitoring, and behavioral analysis requirements of this project and provide the technical foundation necessary to complete this system within one academic term.